PYRAZINAMIDE

Hepatotoxic
Raises uric acid

CHEMICAL NAME Pyrazine-2-carboxamide
CLASS First-line antituberculosis drug (nicotinamide analogue)
TRADE NAMES Zinamide · PZA · Rifater (combination)

THERAPEUTIC USES / INDICATIONS

- Tuberculosis; first 2 months (intensive phase) of HRZE regimen.
- MDR-TB regimens where susceptible; TB meningitis.

PHARMACODYNAMICS

- Converted to pyrazinoic acid in acidic environment → kills semi-dormant bacilli in acidic macrophages and caseous lesions (sterilising effect).
- Allows shortening of TB treatment to 6 months.

ADMINISTRATION

- **PO** once daily, usually in fixed-dose combination tablets.
- With or without food.

DOSAGES

- **Adult:** 25 mg/kg once daily (range 20–30; max 2 g).
- **Children:** 35 mg/kg (30–40) once daily.
- Renal impairment: 25 mg/kg 3 times weekly.

PHARMACOKINETICS

- **Peak:** 2 h. Good CSF penetration.
- Metabolised in liver to pyrazinoic acid (active).
- Excreted in urine. **t½:** 9–10 h.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Nausea, anorexia.
- Arthralgia (joint pain).
- Flushing, rash, photosensitivity.

ADVERSE EFFECTS (SERIOUS)

- Hepatotoxicity (dose-related, most hepatotoxic first-line drug).
- Hyperuricaemia and gout (inhibits uric acid excretion).
- Sideroblastic anaemia; dysuria (rare).

CONTRAINDICATIONS & CAUTIONS

- Severe hepatic impairment; acute gout.
- Porphyria.
- **Caution:** diabetes (harder glucose control), gout history, alcohol, renal impairment.

KEY INTERACTIONS

- **Allopurinol:** may ↑ pyrazinoic acid.
- **Isoniazid, rifampicin:** additive hepatotoxicity.
- **Probenecid:** antagonised.

NURSING MANAGEMENT

- Monitor LFTs and uric acid.
- Ask about joint pain; assess for gout.
- Monitor glucose in diabetics.
- Watch for jaundice, nausea.
- Support adherence during intensive phase.

PATIENT EDUCATION

- Report joint pain, yellow skin or dark urine.
- Avoid alcohol.
- Use sun protection.
- Do not stop TB tablets on your own.

KEY POINTS

- First-line TB drug (Z) for first 2 months.
- Hepatotoxic; raises uric acid (gout).
- Kills dormant bacilli — shortens treatment.
- 25 mg/kg daily.